

Advancements in technology inevitably result in a reduction of human freedom. Do you agree? ^{advocate the expressions freedom of diverse illnesses}

- 📖 The introduction contains an effective thesis that encapsulates the key ideas and arguments of the entire essay.
- ♦ There is a logical development of ideas that reflects good planning which translates into a coherent and succinct discussion.
 - ♦ There is consistent engagement with the key word "inevitably" as the writer firmly denies that a reduction of human freedom is an unavoidable result of technological advancements.

Humans have always sought ways to better their lives. Ever since the Age of Enlightenment and the scientific revolution that accompanied it, humans have become far more proficient at improving their standard of living and quality of life. Observation and deduction – known as the empirical method – allowed man to actualize his dreams. The emergence of science as a discipline produced both technological solutions and products that tackled our problems and provided us with consumer luxuries. Indeed, technology – and by extension, technological advancement – is to be seen as a means to an end. While there have been instances where technological advancements have undeniably invaded privacy and limited freedom of thought, it is difficult to argue that advancements in technology do not fundamentally strive to enable humans to enjoy more freedom and the empowerment that it confers on its possessor. Thus, given my belief that technological advancements ultimately aim to increase human freedom, I disagree with the fallacious claim that it has inescapably constrained our various freedoms.

Advancements in communications technology have commonly been met with suspicion. Technology, that which seemed neutral at first can be used in ways that curtail our freedom more than they empower. Information and communications technology (ICT), hailed to have positively transformed society, has been used (or misused) by agencies to restrict civil liberties. The United States National Security Agency (NSA) surveillance scandal involved the wiretapping and data-mining of over 98.9 million Verizon customers over a three-month period. Through methods that utilize ICT, such as wiretapping and even the hacking of databases, the NSA was reviewed to have violated the freedom, of many, to privacy. In an age of rapid technological advancement, the ability of governments to fashion states and societies that permit and sanction clandestine surveillance programs on a massive and Orwellian scale is indeed troubling. Through surveillance methods, agencies such as the NSA reduce the human freedom to communicate with others freely and without fear. The practice of using products of technological advancement for surveillance purposes in the manner of the NSA is not limited to America, and the electronic snooping of whole communities has become a common practice throughout the world that is conveniently explained away as a necessity for national security. As long as the troubling line between privacy and security is never delineated, it is undeniable that the products of advancements in communication technology will be used in ways that reduce human freedom.

Advancements in technology in areas of information and communication may also be accused of reducing human freedom for another reason. The invention of mobile phones and subsequently smart phones, together with its rapid transition to incorporate computing capabilities, have seen many users in developed nations addicted to these gadgets. It is certainly not difficult to observe this phenomenon in public, where a bulk of individuals

especially those of the younger generation, are seen with their eyes glued to the screens of their phones. The younger generation, in many surveys, have expressed that they feel physical discomfort when they are disallowed from using their mobile devices. Arguably, these products of technological advancements have taken away the ability of individuals to explore other areas of life, chaining many to their electronic gadgets. Indeed, as the capabilities of such products are augmented in the future due to the inexorability of technological progress, one can only posit that Man is unavoidably trading his freedom to explore and engage the real world for a virtual and simulated environment that can only be sought in one's mobile device.

Yet, it is important to recognize that instances where technological advancements lead to a reduction of human freedom typically arise due to the misuse of the technology. Technology is merely a tool that enables and empowers humans. While such tools have undoubtedly been misused before, it is not inevitable that advancements in technology will lead to a reduction in human freedom. Instead, it can be shown in many instances that technological advancements have increased and will continue to increase human freedom. Significantly, technological advancements in medicine have given us freedom from diseases and ailments. Advancements in the areas of biomedical research and biotechnology have given us ways through which we can address and overcome various maladies. The invention of anti-retroviral drugs has allowed individuals suffering from the AIDS to live considerably better lives, providing patients with the freedom to live a life that is almost normal. Recent advancements in genetic technology promise to alleviate inherited and genetic diseases by modifying the genome of individuals suffering from Down syndrome and the Turner syndrome. Advancements in biomedical technology have clearly empowered these individuals to live full and complete lives without being fully dependent on parental support. There are few arguments more compelling than that for biomedical advancements in technology due to the fact that they provide so much freedom and the most fundamental of freedoms – the freedom to live. In view of this, it is difficult to claim that advancements in technology inevitably lead to the reduction of human freedom.

Advancements in technology have also enabled individuals to experience political freedom and fight for their rights. Most notably, ICT tools, specifically social media, were catalysts to mobilise the masses and spread key messages of revolution during the Arab Spring. Used as a means of communicating a unified intention for political revolt and plans for radical social change; revolutionaries convinced and verbalized through social media platforms the need for enlightened societies premised on egalitarian and democratic systems. In countries such as Egypt and Libya, where dissent or the freedom to express one's views were curtailed and censored, social media enabled an outpouring of that same freedom of expression that had been repressed by their respective authoritarian regimes. These examples highlight how ICT, only possible due to advancements in technology in those areas, acted as an agent of change, conferring on the everyman the power and freedom to freely communicate and influence public opinion. Tipping the balance of power of information dissemination in favour of the man-in-the-street has, in this case augmented rather than constrained human freedom.

Advancements in Internet technologies can be said to have greatly contributed to the freedom of expression of people. The Internet has allowed us to communicate our ideas freely and anonymously whenever necessary. Previously marginalized groups now have an avenue to voice their concerns, and gain awareness of their cause. Furthermore, for the average netizen, the cloak of anonymity has enabled many who would not be courageous

enough to give their opinions in public to share constructive criticism. Through avenues such as blogs and social media, individuals are allowed to communicate anything that they want with little government interference. It is thus undeniable that advancements in technology have greatly increased freedom of expression in countries all over the world.

History has shown us how advancements in technology can bring about both an increase in and decrease of human freedom. Yet, such advancements in technology are not intrinsically constructed to protect or take away freedom. Rather, it is imperative to see how these technological aids are used and to what end they are used for. In its potential for both misuse and use, it is impossible to claim with much certainty that a reduction of human freedom would be the inevitable consequence of technological advancement. Given that humans ultimately seek to better their lives and the largely valid assumption that humans ultimately wish to be free to do the things that they desire, it is unlikely that advancements in technology would be allowed to take away more than it provides us with. This is due to the fact that the scientists, engineers and programmers would naturally seek ways to create technologies that will empower humans and improve their lives. Therefore, while I accept the view that technological advancements can reduce human freedom and have done so in the past, I believe that it is fallacious to make the generalization that all advancements in technology would lead us inevitably to a reduction in human freedom. Instead, it is highly likely, given the current trajectory of technological development and its potential for augmenting human freedom, that it will continue to empower humanity.

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